Forecasting Time-Series Data in Power BI

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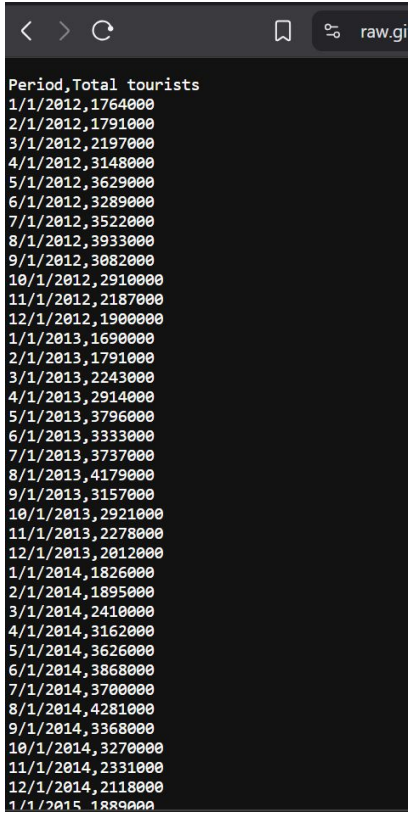
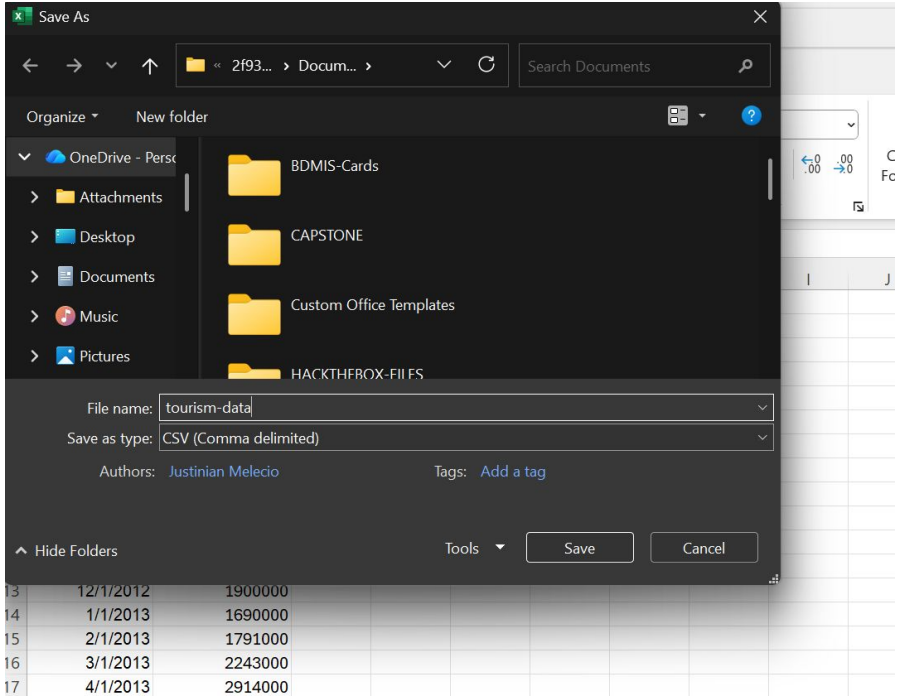
IS 107 - Business Intelligence

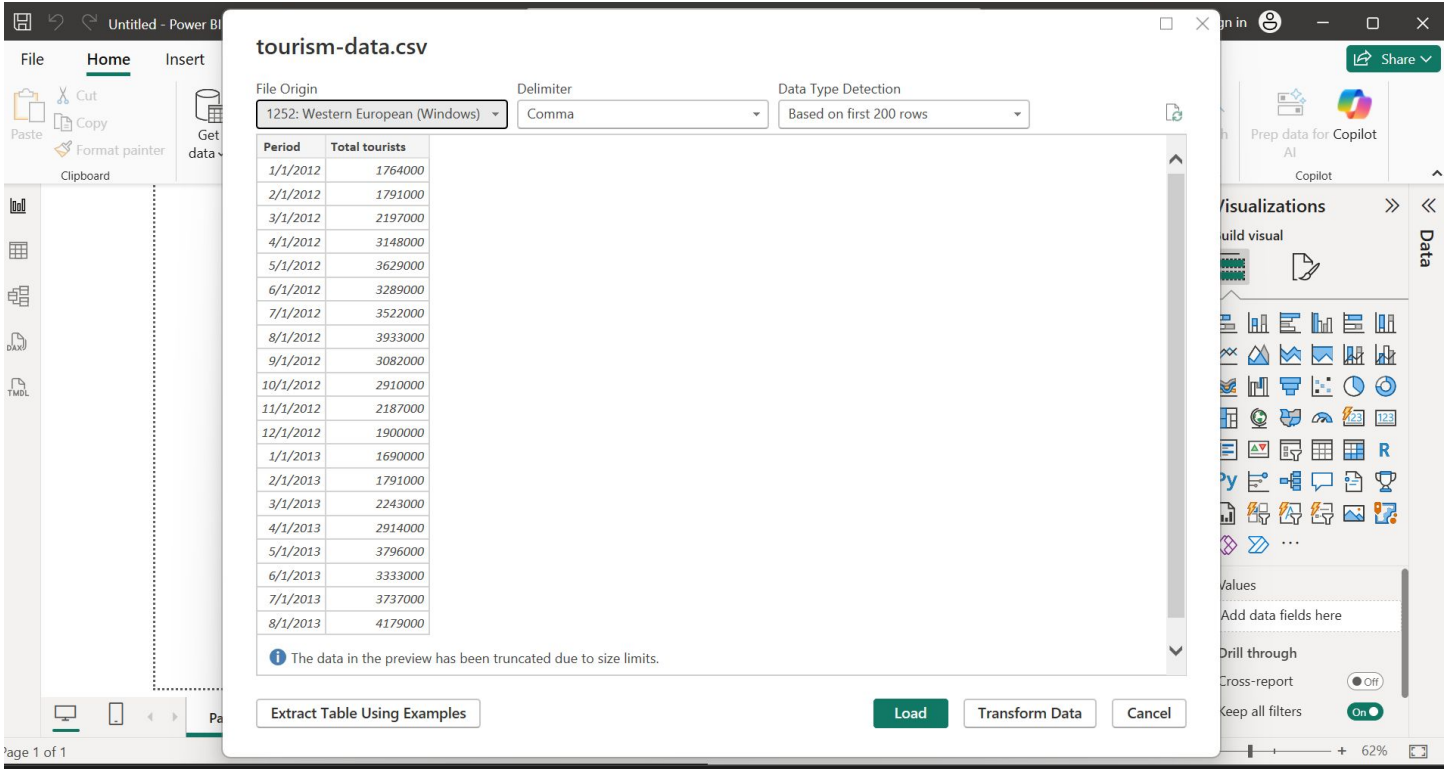
Part 1| Step 1: Replicating the Tourism Data Forecast





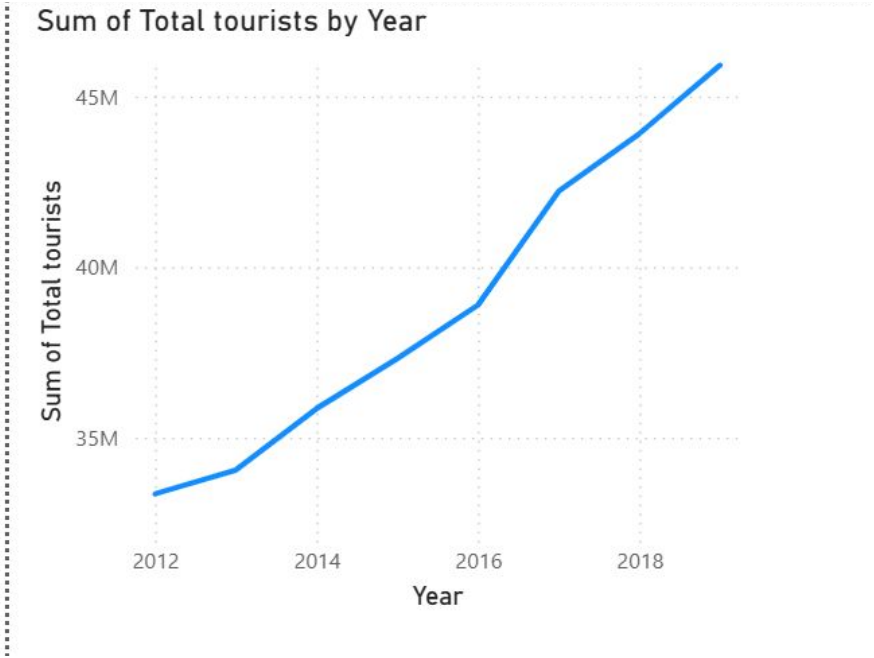
Saving the csv file into computer

Part 1| Step 2: Import Data into Power BI

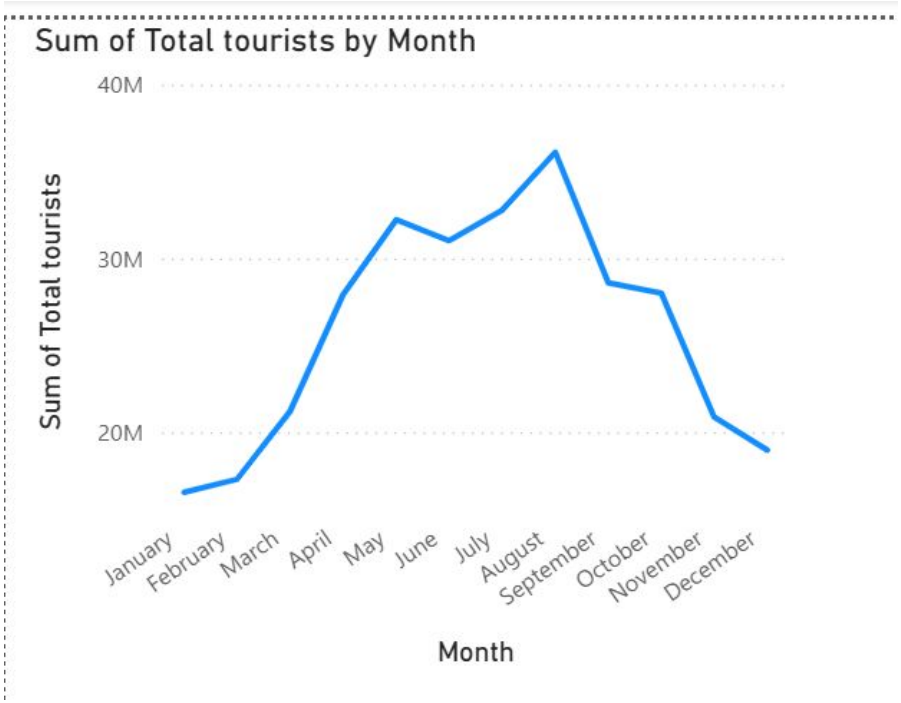


Importing data to Power BI through GET DATA

Part 1|Step 3: Create a Line Chart for Yearly Data | Line Chart

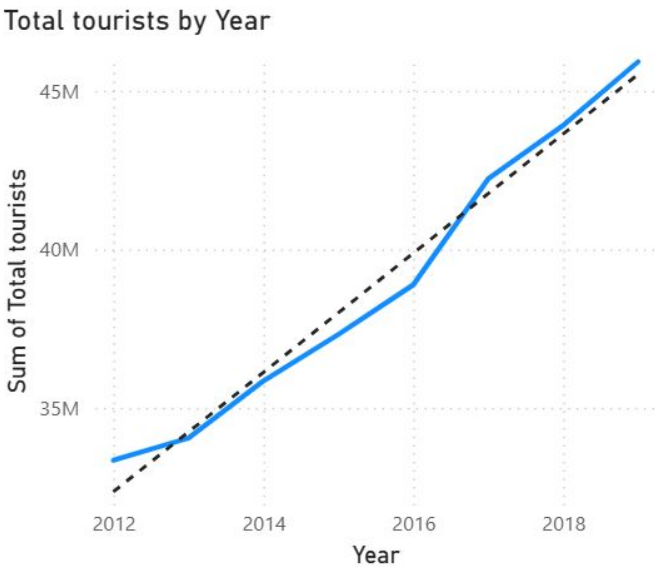


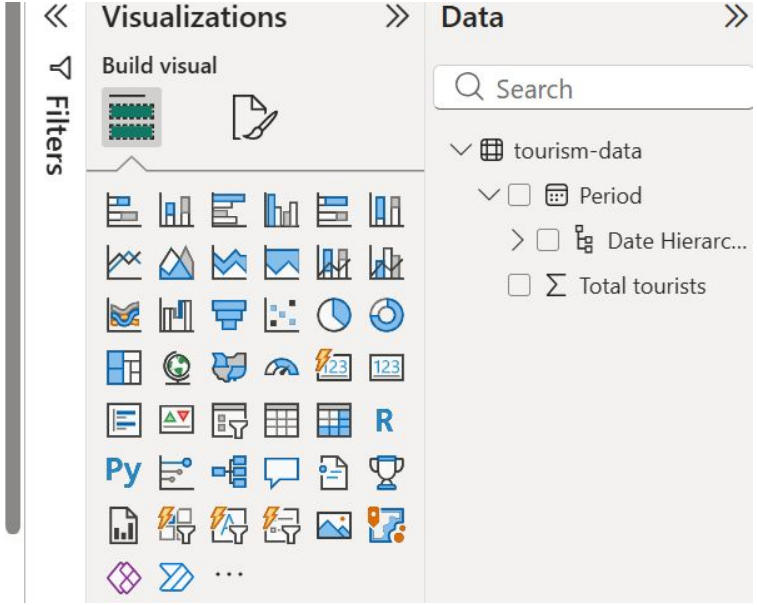
Line Chart for the Sum of tourists by Year hierarchy



Line Chart for the Sum of tourists by Month hierarchy

Part 1|Step 4: Add a Trend Line



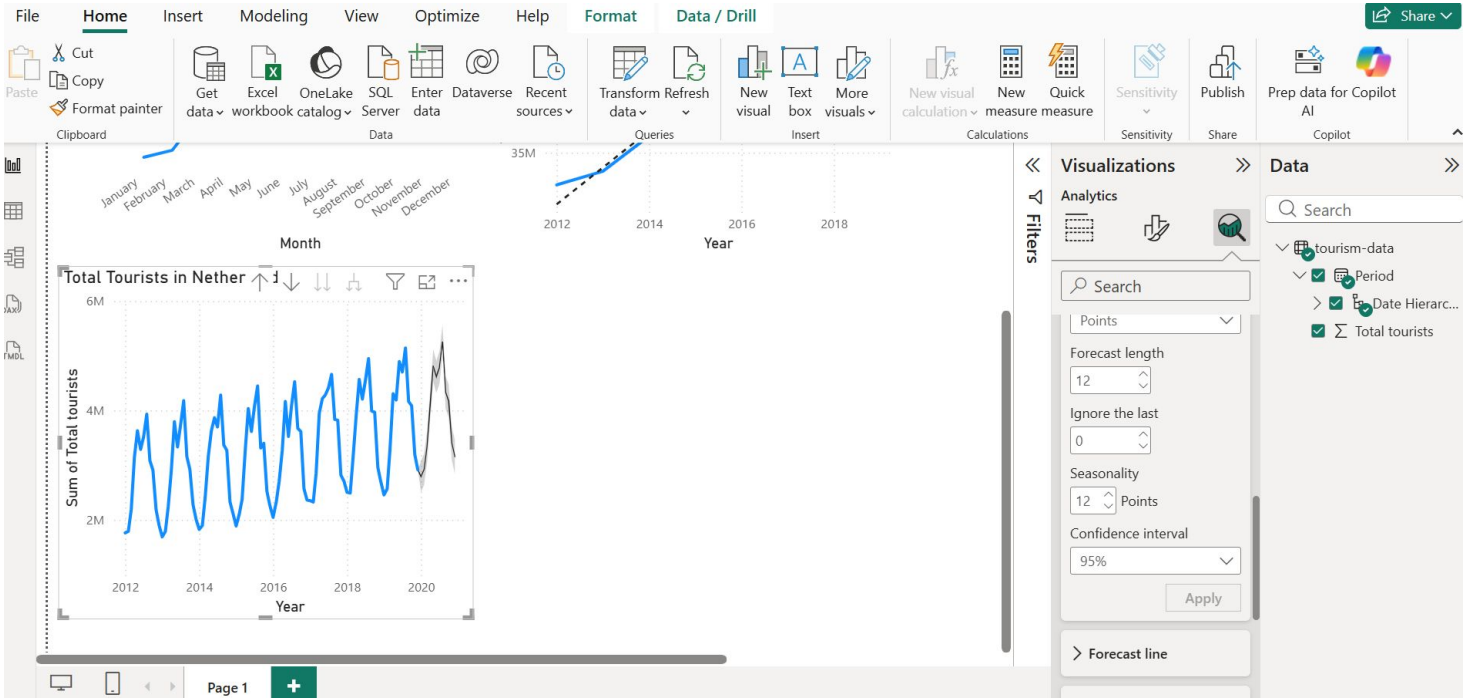


Line Chart for the total tourists with trend line

Part 1|Step 5: Add Forecasting

Line Chart for the Sum of tourists with forecasting analysis

Part 1|Step 6: Configure the Forecast



Line Chart for the Sum of tourists with forecasting analysis that is configured to specific points

Part 1|Step 7: Analyze the Forecast

1. Observe the generated forecast for the year 2020.

- The forecast for 2020 shows a sharp decline in tourist numbers, followed by some recovery afterward. This reflects a major disruption (likely an external event), causing values to drop far below the usual seasonal pattern seen in previous years.

2. Note the confidence interval (the shaded gray area).

What does its narrowness suggest about the forecast's accuracy for this period?

- The wide gray confidence interval shows low accuracy and high uncertainty, especially after 2020, because the sudden drop disrupted the usual pattern and made predictions harder.

3. Compare the forecast to the actual data for 2019.

- The 2019 data shows a stable seasonal pattern, but the forecast after it becomes irregular and less reliable since the model could not handle the sudden change in 2020.

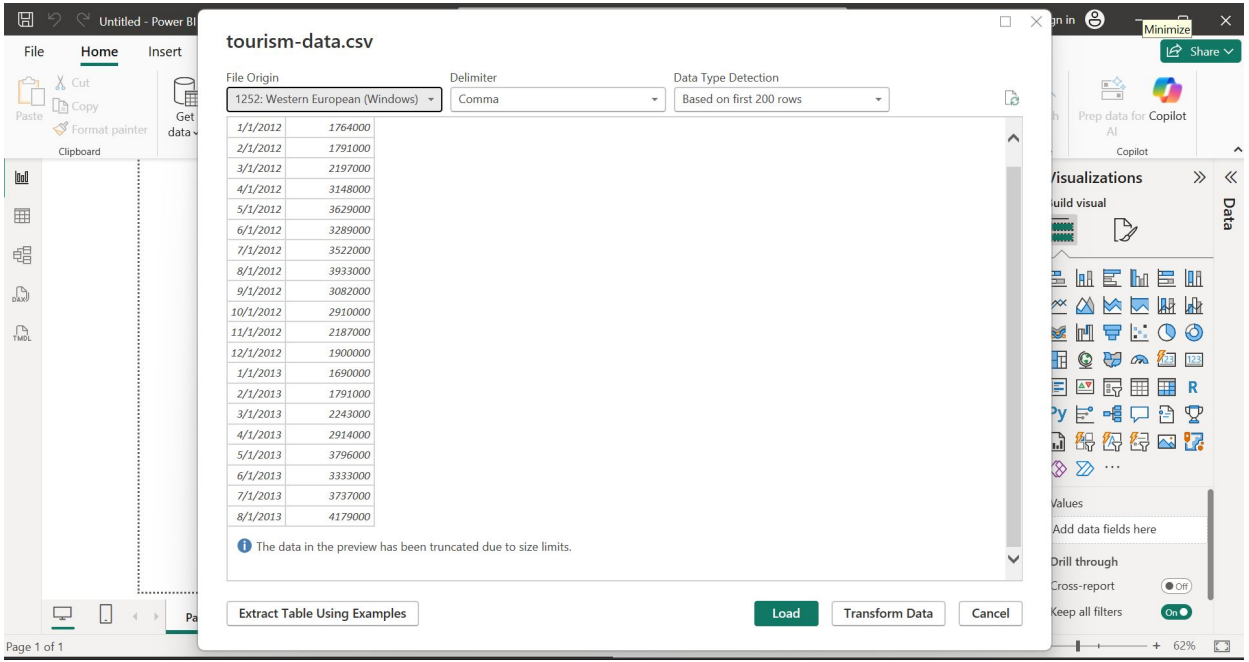
Part 2 Enhancement |Step 8: Prepare a New Dataset

	A	B	C
80	7/1/2018	4546000	
81	8/1/2018	4947000	
82	9/1/2018	3991000	
83	10/1/2018	3967000	
84	11/1/2018	2959000	
85	12/1/2018	2692000	
86	1/1/2019	2458000	
87	2/1/2019	2565000	
88	3/1/2019	3289000	
89	4/1/2019	4307000	
90	5/1/2019	4193000	
91	6/1/2019	4894000	
92	7/1/2019	4704000	
93	8/1/2019	5143000	
94	9/1/2019	4165000	
95	10/1/2019	4087000	
96	11/1/2019	3194000	
97	12/1/2019	2918000	
98			

	A	B	C
130	9/1/2022	3900000	
131	10/1/2022	4100000	
132	11/1/2022	4200000	
133	12/1/2022	4300000	
134	1/1/2023	4400000	
135	2/1/2023	4500000	
136	3/1/2023	4700000	
137	4/1/2023	4900000	
138	5/1/2023	5100000	
139	6/1/2023	5000000	
140	7/1/2023	5200000	
141	8/1/2023	5400000	
142	9/1/2023	5300000	
143	10/1/2023	5500000	
144	11/1/2023	5600000	
145	12/1/2023	5700000	
146	1/1/2024	5800000	
147	2/1/2024	5900000	

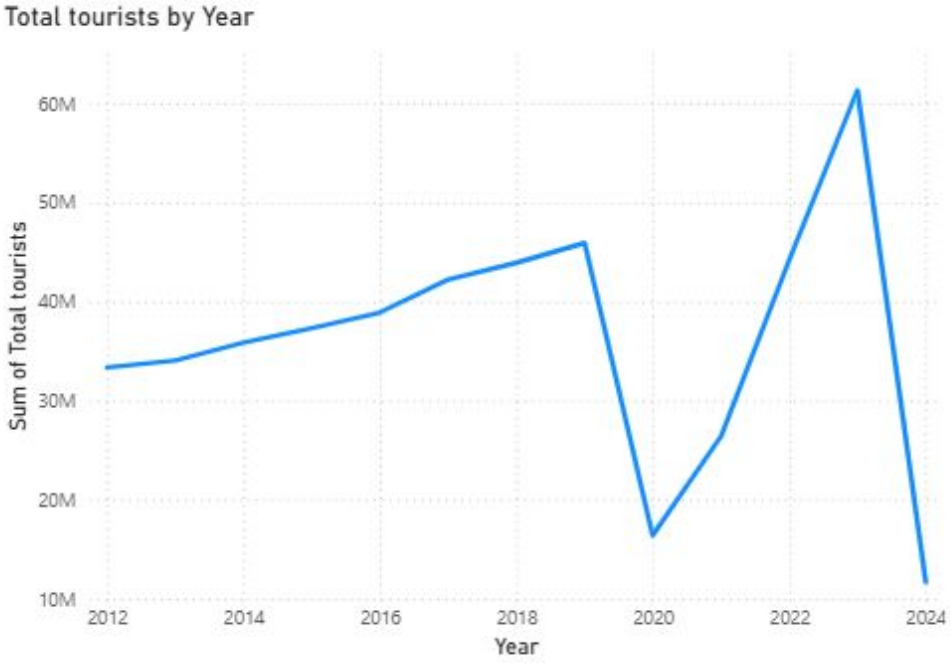
Added another 50 set of data for modification and extra disruptions

Part 2 Enhancement |Step 9: Import and Prepare Your New Dataset



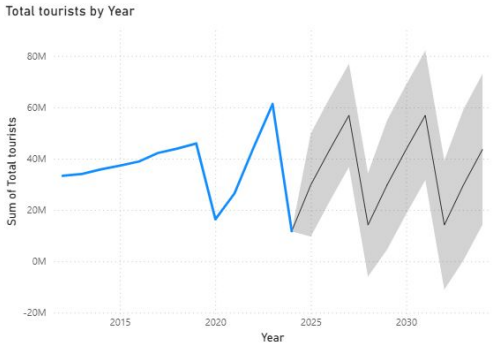
Loading the modified data set by using get data

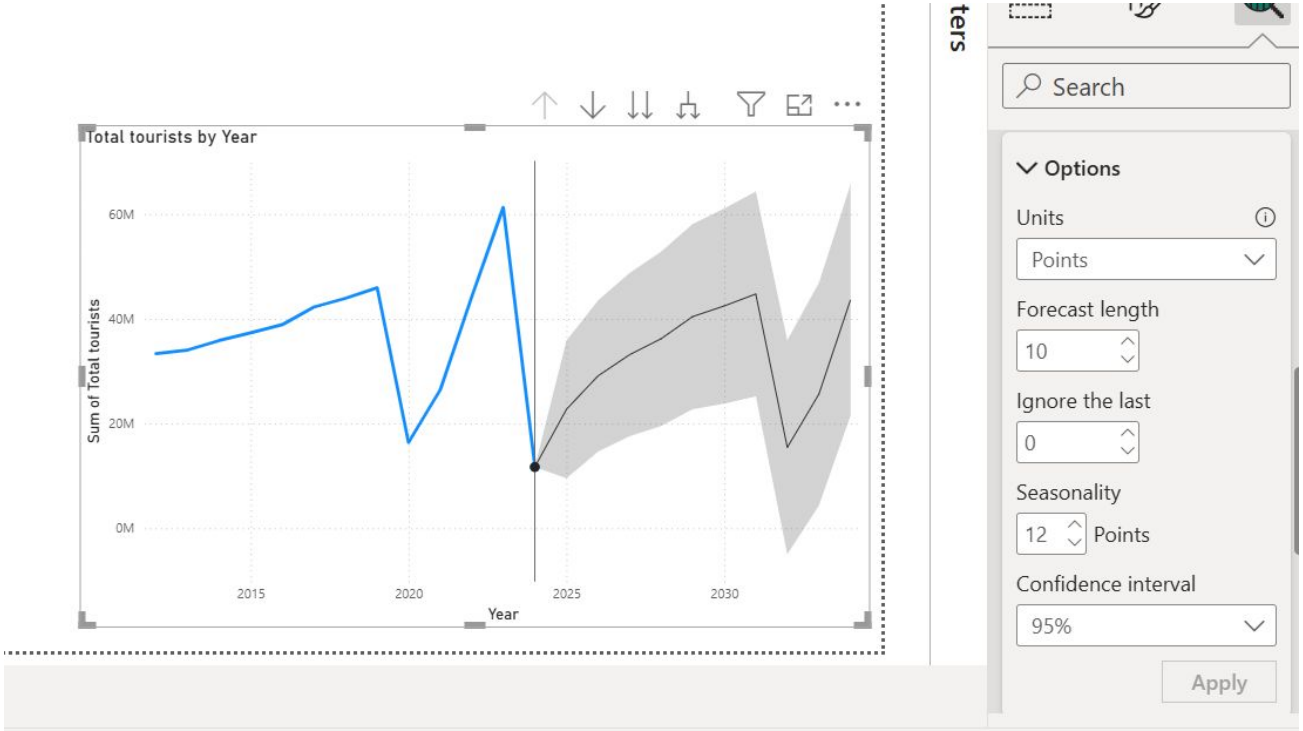
Part 2 Enhancement | Step 10: Create a Line Chart for Your Data



Line Chart for the Sum of tourists by Year hierarchy

Part 2 Enhancement | Step 11: Add and Configure the Forecast





Left figure as the original and the right figure as the configured data forecasting by changing the points analysis

Part 2 Enhancement | Step 12: Analyze and Report Your Findings

1. Describe your dataset:

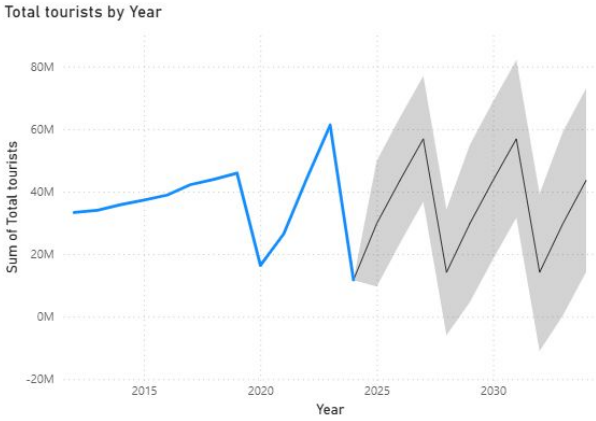
- The dataset contains monthly tourist arrivals from 2012 to 2024, combining historical data (2012–2019) with added disrupted and recovery data (2020–2024). It represents time-series data with seasonal patterns and a major drop during 2020, simulating real-world disruptions.

2. Visualization of Forecast:

- A Power BI line chart was used to visualize the total tourists by year, with the forecasting feature applied. The chart on the right shows historical data as a blue line and the forecast as a continuation with a shaded gray confidence interval.

3. Interpretation of the Forecast:

- The data shows an upward trend with strong seasonality before 2020. The forecast continues this trend but becomes uncertain after the disruption, as shown by the wide confidence interval. Adjusting settings like “Ignore last” smooths drops, while changing the confidence interval affects certainty. The model did not predict the disruption and instead showed higher uncertainty.

4. Conclusion:

- Power BI’s forecasting is useful for identifying trends and seasonality in stable data, but it has limitations when handling sudden disruptions. Its strength lies in simplicity and quick insights, while its weakness is reduced accuracy when unexpected events significantly alter patterns.